Cake shop problem report

1.Problem Description

The proprietor of a cake shop wants to determine the optimal number of 10-inch white birthday cakes to produce each day in order to maximize profit. Daily demand for cakes is uncertain and varies randomly. Producing fewer cakes than demanded leads to lost sales, while producing more cakes results in unsold cakes that are sold at a salvage value or represent a loss. Currently, the shop owner relies on personal judgment to estimate demand, which is inefficient. Therefore, a Monte Carlo Simulation approach is used to model daily demand uncertainty and evaluate different production quantities.

2.Objective

The main objective of this simulation is: To determine the daily production quantity that yields the maximum expected profit.

3.Model Parameters

The following parameters are used in the simulation:

Parameter	Description	Value
P	Selling price	8.5
C	production cost / cake	5
S	Salvage	2

4.Profit Calculation

The profit for each day is calculated based on the relationship between production quantity (x) and demand (d):

- **If production exceeds demand (x > d):**

Profit = $d(p - c) + (x - d)(s - c)$

- **If production is less than or equal to demand ($x \leq d$):**

Profit = $x(p - c)$

5.Simulation procedure

1. Production quantities tested: 50, 60, 70, and 80 cakes.
2. For each quantity, the simulation runs for 10 days.
3. A random number is generated each day to determine demand.
4. Daily profit is calculated and accumulated.
5. Total profit for each production quantity is recorded.
6. The quantity with the highest total profit is selected as optimal.

6.Results

The simulation outputs the total profit for each tested production quantity. The maximum profit obtained determines the best production decision. The simulation results indicate that:

The optimal production quantity is the one that yields the maximum total profit over the simulation period.

7.Conclusion

Using Monte Carlo Simulation allows the cake shop owner to make data-driven decisions instead of relying on guesswork. By simulating random daily demand and calculating profits, the optimal production quantity can be identified, leading to higher expected profits and reduced losses. This approach can be extended by increasing the number of simulated days or modifying demand distributions for greater accuracy.

8.Excel Tabel

D.deman	relativeFr	CDF	RNR
0	0.05	0.05	0 -> 5
10	0.1	0.15	6 -> 15
20	0.05	0.2	16 -> 20
30	0.1	0.3	21 -> 30
40	0.15	0.45	31 -> 45
50	0.3	0.75	46 ->75
60	0.15	0.9	76 -> 90
70	0.05	0.95	91 -> 95
80	0.05	1	96 -> 100

day	randnum	D.deman	profq=50	profQ=60	profQ=70	profQ=80
1	0.94	70	245	270	245	220
2	0.69	50	175	150	125	100
3	0.99	80	355	330	305	280
4	0.65	50	175	150	125	100
5	0.92	70	295	270	245	220
6	0.1	80	355	330	305	280
7	0.06	10	-65	-90	-115	-140
8	0.45	40	115	90	65	40
9	0.49	50	175	150	125	100
10	0.47	50	175	150	125	100
total profit	total profit	total profit	2000	1800	1550	1300

quntity	profit
50	2000
60	1800
70	1550
80	1300
max	2000

The project code

```
#include<iostream>
#include<stdlib.h>
#include<time.h>
#include<cmath>
using namespace std;
double RandNum(double low, double high)
{
	double r = (double)rand() / (double)(RAND_MAX + 1);
	double rawVal (low + (r * (high - low)));
	return round(rawVal * 100.0) / 100.0;
}

int main()
{
	double p = 8.5, c = 5, s = 2.5, r;
	double z = 0, total_z = 0;
	int x, d, day;
	double maxx = 0;
	int idx = 0;
	int daily_dem[9] = { 0,10,20,30,40,50,60,70,80 };
	double cdf[10] = { 0,0.05  , 0.15 , 0.2 , 0.3 , 0.45 , 0.75 , 0.9 , 0.95 , 1 };
	srand((unsigned)time(0));
	for (x = 50; x <= 80; x += 10)
	{
		total_z = 0;
		for (day = 1; day <= 10; day++)
		{
			r = RandNum(0, 1);
			//cout<<"The random number is :" << r << endl;
			for (int i = 0; i < 10; i++)
			{
				if (r >= cdf[i] && r < cdf[i + 1])
				{
					d = daily_dem[i];

					break;
				}
			}
```

```
			if (x > d)
			{
				z = (6 * d) - (2.5 * x);  //z = d(p-c) + (x-d)(s-c)
			}
			else if(x>=d)
			{
				z = 3.5 * x;		// z = (p-c)x
			}
			total_z += z;

		}
		cout << "The total_profit when the quatitiy equal " << x << " is :" << total_z << 
endl;

		if (total_z > maxx)
		{
			maxx = total_z;
			idx = x;
		}

	}
	cout << "The max profit is :" << maxx << " At quantity equal " << idx << endl;

	return 0;
}
```